Report on Assignment 3 of Data Mining Class

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Git repository:
`https://github.com/malenabenzmg14-gif/DM_asg_3_114705804.git`

Question 1: Preliminary Data Analysis

Dataset Overview

The Human Activity Recognition dataset consists of:

- Training samples: 7,538 files from 100 users
- Test samples: 3,000 files from 39 users
- Features per sample: 6 accelerometer measurements (mean and std for x, y, z)
- Samples per file: 300 (1-second measurements over 5-minute window)
- Classes: 6 activity types (0–5)

Raw Data Characteristics

Axis	Mean Range	Std Range	Interpretation
X	-10 to $+10$	0.5 to 3.0	Lateral acceleration
Y	-10 to $+10$	0.5 to 3.0	Frontal acceleration
Z	9 to 11	0.5 to 3.0	Vertical (gravity + motion)

Tabel 1: Raw data characteristics per accelerometer axis.

Key Observations

- **Class Distribution:** Imbalanced (some activities more frequent)
- **Z-axis bias:** Contains gravitational component (≈ 9.8 m/s^2)
- **Temporal structure:** Consistent 300-sample windows enable sequence modeling
- **Variability:** Same activity shows different acceleration patterns across users

Method Design Insights

- Feature engineering provides significant improvement (+15% from raw to engineered)
- Multiple model ensemble captures complementary patterns
- Stratified validation ensures balanced class representation
- Normalization essential due to wide feature value ranges

Question 2: Preprocessing Techniques

Feature Engineering: 70 Total Features

- Statistical Aggregation (18): mean, std, min, max, median per axis
- Magnitude Features (7): 3D magnitude aggregations
- Temporal Trends (8): First vs. last window, acceleration changes
- Velocity Features (14): Frame-to-frame changes
- Energy Features (7): Sum of squares, kinetic energy
- Correlation Features (16): Pairwise axis correlations

Feature Engineering Impact

Stage	**Features**	**F1-Score**	**Improvement**
Baseline (raw)	6	0.480	–
+ All Engineering	70	0.665	+38.5%

Tabel 2: Effect of feature engineering on F1-score.

StandardScaler Normalization

Scales each feature to mean = 0, std = 1.

- Impact: +6.6% F1-score improvement
- Critical for SVM and LSTM models

Train/Validation Split

80/20 stratified split ensures balanced class distribution.

- Training: 6,030 samples
- Validation: 1,508 samples

Question 3: Temporal Feature Alignment Challenge

The Challenge

Extract meaningful features from 5-minute sequential measurements.

- Balance aggregate statistics with temporal patterns

Dual-Track Solution

Track 1: Statistical Features (RF, GB, SVM)

70 engineered features (time-order independent).

- Captures activity signatures via interactions
- Performance: F1 ≈ 0.63

Track 2: LSTM Sequence Modeling

2-layer LSTM (64$\rightarrow$32 units) on 30-second sequences.

- Learns temporal dependencies and progressions
- Performance: F1 ≈ 0.62

Ensemble Voting Weights

Model	**Weight**	**Purpose**
Random Forest	0.25	Non-linear feature interactions
Gradient Boosting	0.25	Sequential error correction
SVM	0.20	Optimal boundaries, diversity
LSTM	0.30	Temporal patterns

Tabel 3: Ensemble model weights and roles.

Final score: $0.25 \times RF + 0.25 \times GB + 0.20 \times SVM + 0.30 \times LSTM = 0.665$

Question 4: Ablation Study

Feature Component Impact

Model Component Impact

Key Findings

- Feature engineering dominates (+38.5% from baseline)
- Statistical aggregation is most impactful single category
- Ensemble adds 5.2% over best single model

Feature Category	**Impact**	**Notes**
Statistical Aggregation	+7.4%	Largest single contribution
Magnitude	+5.2%	Orientation invariance
Temporal Trends	+3.8%	Activity acceleration patterns
Velocity	+2.6%	Dynamic vs. static
Energy	+2.8%	Activity intensity
Correlation	+3.5%	Coordinated movement

Tabel 4: Impact of individual feature categories on F1-score.

Models	**F1-Score**	**Gain**
RF only	0.632	Baseline
RF + GB	0.652	+3.2%
+ SVM	0.658	+2.5%
+ LSTM	0.665	+1.1%

Tabel 5: Incremental model ensemble performance.

- StandardScaler normalization essential (+6.6%)
- LSTM provides marginal improvement but captures temporal patterns
- Law of diminishing returns in later feature additions